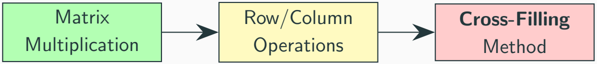
Cross-Filling Method

Finding Hidden Patterns in Recipe Tables

Introduction

Learning Objectives



Key Questions:

- Can we **discover hidden intermediate products** from a direct recipe table?
- What is a **simple pattern** (rank-one table)?
- How to **decompose** complex tables into simple patterns?
- How to convert between **sum** and **product** forms?

Review: Sum-of-Rank-One View

Recall from Lecture 1, matrix multiplication $C = AB$ has three views:

$$C = \begin{pmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} - & b_1^T & - \\ - & b_2^T & - \\ \vdots & \vdots & \vdots \\ - & b_n^T & - \end{pmatrix} = \sum_{k=1}^{n} a_k \, b_k^T$$

Each $a_k \, b_k^T$ is called a **rank-one piece** (column $\times$ row).

Today's question: Can we **reverse** this? Given C, find the pieces?

Discovering Simple Patterns (Concretization)

Coffee shop scenario:

We have a **direct** recipe table from raw materials to final products:

$C =$

	🍱	🍲
🍃	2	2
🍋	1	1
	0	4
🐄	2	1

Question: Are there **hidden intermediate products** (like 🥛, ☕, 🍵) that we could use to simplify this?

What is a Simple Pattern?

Look at the first two rows of our table:

	🍱	🍲
🍃	2	2
🍋	1	1

Notice: Both products use **the same amount**!

- 🍱: 2🍃 + 1🍋
- 🍲: 2🍃 + 1🍋

Can we extract this **common base** and see what's different?

Setup: The Full Direct Table

$C =$

	🍱	🍲
🍃	2	2
🍋	1	1
🫘	0	4
🐄	2	1

We noticed: first two rows have the same pattern.

Can we **extract** this pattern and see what's left?

Change: Choose a Pivot

Pick a pivot (center of our cross):

Let's pick the 🍃 row and 🍱 column. Entry $= 2$.

$C =$

	🍱	🍲
🍃	**2**	2
🍋	1	1
	0	4
🐄	2	1

This pivot will be the **center of our cross**.

Calculation Step 1: Extract the Cross

The **cross** through the pivot includes:

- Entire 🍃 row (row through pivot)
- Entire 🍱 column (column through pivot)

$C =$

	🍱	🍲
🍃	**2**	2
🍋	1	1
🫘	0	4
🐄	2	1

Cross row: $(2, 2)$ Cross column: $(2, 1, 0, 2)$

Calculation Step 2: Build the Pattern Table

Idea: Use the cross to fill all entries via "multiplication table":

$$\text{New entry} = \frac{(\text{column value}) \times (\text{row value})}{\text{pivot}}$$

For  and :

$$\text{Entry} = \frac{1 \times 2}{\mathbf{2}} = 1$$

For  and :

$$\text{Entry} = \frac{0 \times 2}{\mathbf{2}} = 0$$

Result: The First Pattern R_1

$R_1 =$

	🍱	🍲
🍃	2	2
🍋	1	1
	0	0
🐄	2	2

Interpretation: This is the **common base recipe**!

- Base recipe: 2🍃 + 1🍋 + 0 + 2🐄
- 🍱 uses: 1× base
- 🍲 uses: 1× base (+ adjustments)

Insight: What is This Pattern?

The pattern R_1 is a **rank-one matrix**:

$$R_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix}$$

- **Column** $(2, 1, 0, 2)$: the base recipe - **Row** $(1, 1)$: both products use $1\times$ this base

Key: **One pattern, repeated!** Both columns are the same.

Setup: What's Left After R_1?

Subtract R_1 from the original table C:

$C - R_1 =$

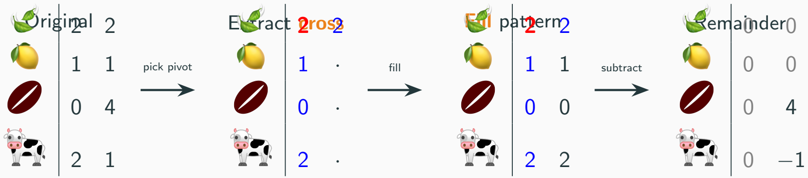
	🍱	🍲
🍃	$2-2$	$2-2$
🍋	$1-1$	$1-1$
🫘	$0-0$	$4-0$
🐄	$2-2$	$1-2$

$C_2 = C - R_1 =$

	🍱	🍲
🍃	0	0
🍋	0	0
🫘	0	4
🐄	0	-1

What's left? - 🍱 column: all zeros (base recipe exactly matches!) - 🍲 column: adjustments needed (+4 🫘, -1 🐄)

Insight: Why "Cross-Filling"?



The method **fills in** the cross region, then eliminates it!

Look at the remainder:

$$C_2 = \begin{array}{c|cc} & \text{🍱} & \text{🍲} \\ \hline \text{🫘} & 0 & 4 \\ \text{🐄} & 0 & -1 \end{array}$$

This is the **adjustment** for 🍲: $+4$ 🫘, -1 🐄

Pick pivot: 🫘 row, 🍲 column. Entry $= 4$.

Calculation: Fill Second Cross

Pivot at (3,2) $= 4$. Cross: column $(0, 0, 4, -1)^T$, row $(0, 4)$

$$R_2 = \frac{1}{\mathbf{4}} \begin{pmatrix} 0 \\ 0 \\ 4 \\ -1 \end{pmatrix} \begin{pmatrix} 0 & 4 \end{pmatrix} =$$

	🍱	🍲
🍃	0	0
🍋	0	0
	0	**4**
🐄	0	-1

Interpretation: The **adjustment term** — only affects 🍲.

Result: Complete Decomposition

$C_2 - R_2 =$

	🍱	🍲
	0	0
🐄	0	0

Done! Remainder is zero.

The Full Decomposition

We discovered:

$$C = R_1 + R_2$$

$$\begin{array}{c|cc} \text{🍃} & 2 & 2 \\ \text{🍋} & 1 & 1 \\ \text{🫘} & 0 & 4 \\ \text{🐄} & 2 & 1 \end{array} = \begin{array}{c|cc} \text{🍃} & 2 & 2 \\ \text{🍋} & 1 & 1 \\ \text{🫘} & 0 & 0 \\ \text{🐄} & 2 & 2 \end{array} + \begin{array}{c|cc} \text{🍃} & 0 & 0 \\ \text{🍋} & 0 & 0 \\ \text{🫘} & 0 & 4 \\ \text{🐄} & 0 & -1 \end{array}$$

Two **rank-one patterns**: base recipe + adjustment!

Insight: Base + Adjustment Decomposition

What we found:

🍲 $= 1 \times (\text{base recipe}) + \text{adjustment}$

Key insight:

- R_1: Common base shared by both products
- R_2: Adjustment term (only for 🍲)
- 🍱 = exactly the base recipe
- 🍲 = base + extra 4 🫘 - 1 🐄

This is **expressing complex recipes in terms of simpler patterns**!

Change of Perspective

Change of Perspective

What we've learned so far:

- **What** a simple pattern (rank-one) looks like — one base mix used in different amounts
- **How** to extract patterns using cross-filling — pick pivot, fill cross, subtract
- **Why** this works — discovers hidden intermediate products

From This Point Forward

We **forget about coffee shops** and work with **abstract matrices**:

- **What**: Matrices A, B, C, rank-one pieces R_k
- **How**: Cross-filling algorithm, sum $\leftrightarrow$ product conversion
- **Why**: Compute rank, factorizations, solve equations

This simplifies notation while preserving the intuition.

Rank-One Matrices (Formalization)

Definition: Rank-One Matrix

Definition (Rank-One Matrix)

A matrix R is **rank-one** if it can be written as

$$R = \mathbf{u}\,\mathbf{v}^T$$

for some nonzero column vector $\mathbf{u}$ and nonzero row vector $\mathbf{v}^T$.

Example:

$$R = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 6 \end{pmatrix}$$

Key: All columns are multiples of $\mathbf{u}$. All rows are multiples of $\mathbf{v}^T$.

Recognizing Rank-One Structure

Is this matrix rank-one?

$$\begin{pmatrix} 6 & 9 & 3 \\ 4 & 6 & 2 \\ 10 & 15 & 5 \end{pmatrix}$$

Check rows:

- Row 1: $(6, 9, 3) = 3 \cdot (2, 3, 1)$
- Row 2: $(4, 6, 2) = 2 \cdot (2, 3, 1)$
- Row 3: $(10, 15, 5) = 5 \cdot (2, 3, 1)$

Yes! $R = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix} \begin{pmatrix} 2 & 3 & 1 \end{pmatrix}$

Cross-Product Property

In a rank-one matrix $R = \mathbf{u}\,\mathbf{v}^T$, every entry is $r_{ij} = u_i v_j$.

For any four entries forming a rectangle:

$$\boxed{r_{ij} \cdot r_{k\ell} = r_{i\ell} \cdot r_{kj}}$$

Why? $(u_i v_j)(u_k v_\ell) = (u_i v_\ell)(u_k v_j) = u_i u_k v_j v_\ell$

Check: In $\begin{pmatrix} 6 & 9 \\ 4 & 6 \end{pmatrix}$: $\quad 6 \times 6 = 36 = 9 \times 4 \quad \checkmark$

Not Every Matrix is Rank-One

$$C = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 7 \end{pmatrix}$$

Check ratios: $\frac{4}{2} = 2, \quad \frac{2}{1} = 2, \quad \frac{7}{3} \neq 2 \quad \leftarrow$ Different!

This is **not rank-one**. But can we write it as a **sum** of rank-one pieces?

$$C = \underbrace{\begin{pmatrix} ? \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} ? \end{pmatrix}}_{R_2} + \cdots$$

Cross-filling answers this!

Cross-Filling Algorithm

The Cross-Filling Algorithm

Proposition (Cross-Filling Algorithm)

Input: Matrix A ($m \times n$)

Output: $A = R_1 + R_2 + \cdots + R_r$ (sum of rank-one matrices)

Procedure:

1. Find any nonzero entry $a_{ij} \neq 0$ (the **pivot**)
2. Extract the cross: row i and column j
3. Form rank-one piece:

$$R = \frac{1}{a_{ij}} \cdot (\text{column } j) \cdot (\text{row } i)$$

4. Compute remainder: $A' = A - R$
5. Repeat on A' until $A' = O$

Each step eliminates one cross (one row and one column become zero).

Example: Cross-Filling a 3×3 Matrix

Decompose:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix}$$

We will peel off rank-one pieces iteratively.

Iteration 1: Pivot $a_{11} = 1$

Highlight the cross:

$$A = \begin{pmatrix} \mathbf{1} & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix}$$

Fill using the cross:

$$R_1 = \frac{1}{\mathbf{1}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \mathbf{1} & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Iteration 1: Remainder A_1

$$A_1 = A - R_1 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 7 \end{pmatrix}$$

Row 1 and column 1 are now zero. ✓

Iteration 2: Pivot $(A_1)_{22} = 2$

$$A_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{2} & 2 \\ 0 & 2 & 7 \end{pmatrix}$$

Fill:

$$R_2 = \frac{1}{\mathbf{2}} \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{2} & 2 \\ 0 & 2 & 2 \end{pmatrix}$$

Iteration 2: Remainder A_2

$$A_2 = A_1 - R_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 7 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Rows 1–2 and columns 1–2 are now zero. ✓

Iteration 3: Final Piece

$$A_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{5} \end{pmatrix}$$

This is already rank-one:

$$R_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{5} \end{pmatrix}$$

Remainder: $A_3 = A_2 - R_3 = O$. **Done!**

Full Decomposition

$$A = \underbrace{\begin{pmatrix} \mathbf{1} & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{2} & 2 \\ 0 & 2 & 2 \end{pmatrix}}_{R_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{5} \end{pmatrix}}_{R_3}$$

Three rank-one pieces:

- R_1: pivot **1** at position $(1, 1)$
- R_2: pivot **2** at position $(2, 2)$
- R_3: pivot **5** at position $(3, 3)$

Definition: Rank of a Matrix

Definition (Rank)

The **rank** of a matrix A is the number of rank-one pieces in its cross-filling decomposition:

$$A = R_1 + R_2 + \cdots + R_r \quad \Longrightarrow \quad \text{rank}(A) = r$$

Examples:

- Zero matrix O: rank 0
- Rank-one matrix $\mathbf{uv}^T$: rank 1
- Identity I_n: rank n
- Our A above: rank 3

Important: Different pivot choices give different decompositions, but always the same count r. (Proved in Lecture 4.)

Sum $\leftrightarrow$ Product Equivalence

From Sum to Product Form

Each rank-one piece is an **outer product**:

$$R_k = \mathbf{u}_k \, \mathbf{v}_k^T$$

From our example:

$$R_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$$

$$R_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 \end{pmatrix}$$

$$R_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 5 \end{pmatrix}$$

Collecting the Pieces

Collect columns $\mathbf{u}_k$ into U, rows $\mathbf{v}_k^T$ into V:

$$U = \begin{pmatrix} | & | & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$V = \begin{pmatrix} - & \mathbf{v}_1^T & - \\ - & \mathbf{v}_2^T & - \\ - & \mathbf{v}_3^T & - \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 5 \end{pmatrix}$$

By sum-of-rank-one: $UV = \sum_{k=1}^{3} \mathbf{u}_k \, \mathbf{v}_k^T = R_1 + R_2 + R_3 = A$

Verification

$$UV = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix} = A \quad \checkmark$$

Notice: U is **lower triangular**, V is **upper triangular**!

This happened because we chose **diagonal pivots** $(1, 1) \to (2, 2) \to (3, 3)$.

Key: Diagonal pivots $\Longrightarrow$ **LU decomposition** (preview of future lecture).

Sum $\leftrightarrow$ Product Equivalence

Proposition (Sum $\leftrightarrow$ Product)

For any matrix A:

$$A = \sum_{k=1}^{r} \mathbf{u}_k \, \mathbf{v}_k^T \quad \Longleftrightarrow \quad A = U \cdot V$$

where U has columns $\mathbf{u}_k$ and V has rows $\mathbf{v}_k^T$.

Two directions:

- **Lecture 1**: Given product $A = UV$, read off sum $A = \sum \mathbf{u}_k \, \mathbf{v}_k^T$
- **Today**: Given sum from cross-filling, assemble product $A = UV$

The Inner Dimension

When $A = UV$ with A being $m \times n$:

$$A_{m\times n} = U_{m\times \boxed{r}} \cdot V_{\boxed{r}\times n}$$

The shared dimension r is the **inner dimension**.

- Outer dimensions m, n: visible (size of A)
- Inner dimension r: **hidden** = number of independent patterns

Key: $\text{rank}(A) = r =$ inner dimension of the factorization.

Algebraic Expression for Cross-Filling

Cross-Filling as a Matrix Formula

Let e_i be the i-th column of I, and e_j^T the j-th row of I.

Then:

- $Ae_i =$ column i of A
- $e_j^T A =$ row j of A
- $e_j^T Ae_i =$ entry (j, i) of $A =$ the pivot

The cross-filling formula at pivot position (row j, column i):

$$\boxed{R = Ae_i \, (e_j^T Ae_i)^{-1} \, e_j^T A}$$

This is: $\dfrac{1}{\text{pivot}} \times (\text{column}) \times (\text{row})$

Example Verification

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix}, \quad \text{cross-fill at row 1, column 1:}$$

$$Ae_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad e_1^T A = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}, \quad e_1^T A e_1 = 1$$

$$R_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot 1 \cdot \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad \checkmark$$

Application: Projection Matrices

Definition (Projection)

A square matrix P is a **projection** if $P^2 = P$.

Proposition

If P is a projection and R is obtained by cross-filling from P, then R is also a projection.

Proof: Let $R = Pe_i(e_j^T Pe_i)^{-1} e_j^T P$. Then

$$R^2 = Pe_i(e_j^T Pe_i)^{-1} \underbrace{e_j^T P \cdot Pe_i}_{= e_j^T Pe_i}(e_j^T Pe_i)^{-1} e_j^T P$$

$$= Pe_i(e_j^T Pe_i)^{-1} e_j^T P = R \quad \checkmark$$

(Used $P^2 = P$ in the underbraced step.)

Projection Decomposition

If P is a projection and R comes from cross-filling P:

$$P = \underbrace{R}_{\text{projection}} + \underbrace{(P - R)}_{\text{also projection!}}$$

Check: $(P - R)^2 = P^2 + R^2 - PR - RP = P + R - R - R = P - R \quad \checkmark$

Cross-filling **decomposes projections into smaller projections**.

This will be **crucial** for spectral decomposition (future lecture).

Preview: LU Decomposition

Preview: Diagonal Pivots ⇒ LU

In our example, diagonal pivots produced:

$$A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}}_{\text{lower triangular } L} \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 5 \end{pmatrix}}_{\text{upper triangular } U}$$

Key: The pivots $\mathbf{1}, \mathbf{2}, \mathbf{5}$ appear on the diagonal of U.

Diagonal cross-filling $\Longrightarrow$ **LU decomposition**.

Preview (Future Lecture)

LU decomposition is a special case of cross-filling with diagonal pivot strategy. Full theory, including PLU decomposition, will be covered in Week 3.

Not All Matrices Have Diagonal LU

Diagonal cross-filling requires **nonzero diagonal pivots**. But this can fail:

$$A = \begin{pmatrix} 0 & 3 \\ 1 & 2 \end{pmatrix}$$

Pivot $a_{11} = 0$ — cannot start!

Even later pivots can fail:

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 1 & 3 & 5 \end{pmatrix} \xrightarrow{\text{after } R_1} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

Second pivot is **0**!

Solution: Allow non-diagonal pivots, or permute rows first → **PLU** (future lecture).

Summary

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Today we learned:

1. **Rank-one matrix**: $R = \mathbf{u}\,\mathbf{v}^T$ (one pattern, repeated)
2. **Cross-filling algorithm**: Decompose any matrix into rank-one pieces

$$A = R_1 + R_2 + \cdots + R_r$$

   Pick pivot → Extract cross → Fill → Subtract → Repeat

3. **Sum ↔ Product equivalence**:

$$\sum_{k=1}^{r} \mathbf{u}_k\,\mathbf{v}_k^T = A = UV$$

4. **Rank** = number of pieces = inner dimension of UV
5. **Algebraic formula**: $R = Ae_i(e_j^T A e_i)^{-1} e_j^T A$
6. **Applications**: Projection decomposition, preview of LU decomposition

Looking Ahead

Lecture 4 (next week):

- **Well-definedness of rank** — Why does rank not depend on pivot choices?
- **Linear independence** and **basis** — Two languages for subspaces
- **Dimension** — Connection to rank via trace
- Solving $Ax = b$ using cross-filling

The big picture:

Cross-filling → Rank → Subspaces → Projections → Spectral Decomposition

Every major topic connects back to rank-one decomposition!